Saurabh Kumar

Google Scholar: tinyurl.com/4pyub4pd
GitHub: github.com/saurabhk0317
Email: saurabhk0317@gmail.com
LinkedIn: saurabhk0317

Education

Indian Institute of Science (IISc) — Bengaluru, India
M.Tech. (Research) in Electrical Engineering — 2026–Present
Advisor: Prof. Prasanta Kumar Ghosh (SPIRE Lab)
GATE 2026 (Data Science and AI): AIR 362/69,242
Publications: [1]

- Proposed a CTC training criterion with token-level wildcard arcs that tolerates localized transcription ambiguity (pronunciation, spelling, lexical variants), achieving a **9.45% average relative WER reduction over CTC** across multiple languages and corpora [1].

National Institute of Technology, Patna — Patna, India
B.Tech. in Electronics and Communication Engineering — 2016–2020
GPA: 7.42/10.00

Experience

SPIRE Lab, Indian Institute of Science (IISc) — Bengaluru, India
Research Associate — July 2021–July 2026
Advisor: Prof. Prasanta Kumar Ghosh
Projects:

- **RESPIN**: *Speech Recognition in Agriculture and Finance for the Poor in India* (funded by the Gates Foundation)
  Publications: [2]–[7]
  - First author of the RESPIN-S1.0 database paper published at **NeurIPS 2025 (Datasets and Benchmarks Track)**; led large-scale data curation, reproducible benchmarking, and baseline ASR development across nine Indian languages [2].
  - Developed a multimodal fusion model for joint ASR and dialect identification (DID) (**4.65% CER and 81.63% DID accuracy**), integrating speech and text embeddings for Indian dialects [4].
  - Led the first large-scale study on DID in Indian languages (**79.81% accuracy across 33 dialects**) [5].
  - Enhanced low-resource ASR via adapter-based domain adaptation, reducing **WER (10%-30%)** and **CER (3%-18%)** across 9 languages [6].
  - Proposed a Gated Multi-Encoder ASR model with frame-level gating, reducing **WER by 7.5% (Bhojpuri) and 9% (Bengali)** [7].
- **SYSPIN**: *Text-to-Speech Synthesizer in Nine Indian Languages* (funded by GIZ, Germany)
  Publications: [8]
  - Developed a data filtering pipeline using Kaldi-based ASR models, improving corpus quality.

ARTPARK @ IISc — Bengaluru, India
Machine Learning/Signal Processing Consulting Engineer — Sep 2022–July 2026
Project:

- **VAANI**: *Capturing the language landscape for an inclusive digital India* (funded by Google)
  Publications: [9]

- Developed automated signal processing and machine learning pipelines for acoustic quality assessment, speech activity detection, and metadata validation.
- Established quantitative acceptance criteria and automated corpus curation protocols underpinning the open-source release of **31,000+ hours** of speech from **156,000+ speakers** across **109 languages** and **31 states and union territories**.

Department of ECE, NIT Patna Patna, India
Undergraduate Student Researcher July 2019–June 2021
Advisor: Dr. S. Shahnawazuddin
Publications: [10], [11]

- Investigated formant-modification and bispectrum-based front-end features to improve children's ASR under zero-resource conditions [10].
- Developed a CGAN-based voice conversion framework, transforming adult speech into child-like speech for ASR adaptation [11].

EICT Academy, IIT Guwahati Guwahati, India
Summer Intern May 2019–June 2019

- Designed Verilog code for a 5-stage discrete wavelet transform cascade filter bank for FPGA-based simulation.

Mentorship & Professional Services

Reviewer:

- **Conferences and Workshops:** ICASSP (since 2025), IJCNN (since 2025), IEEE ASRU 2025, Interspeech (since 2026), IEEE SLT 2026.
- **Journals:** Computer Speech & Language (Elsevier), Speech Communication (Elsevier).

Organizer:

- **ASRU 2025 (MADASR 2.0):** Lead organizer of the Multi-Lingual Multi-Dialect ASR Challenge in 8 Indian Languages; designed challenge tracks, evaluation protocols, public leaderboards, developed ESPnet-based baseline systems, and reviewed challenge track submissions.
- **ICASSP 2024 (LIMMITS'24):** Co-organized session on Indic TTS, developed baselines, and reviewed challenge track submissions.
- **ASRU 2023 (MADASR):** Led data curation, ASR baselines for Bengali, Bhojpuri.
- **Interspeech 2022 (Gram Vaani ASR):** Organized Hindi ASR challenge, developed Kaldi baselines.

Student Volunteer: ICASSP 2024, ICASSP 2025.

Awards

SLT-CODE Hackathon Winner 2022 – Won **Best Hackathon Project** Award for developing dialectal ASR systems for Bengali and Bhojpuri, presented at **SLT 2022**.
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Languages and Skills

- **Programming:** Python, Bash
- **ML/DL Toolkits:** PyTorch, Scikit-learn
- **ASR Frameworks:** ESPnet, Kaldi, Fairseq
- **Tools:** MATLAB, Audacity, Praat, Git, Hugging Face, Linux, Docker, LaTeX
- **Languages Spoken:** Bhojpuri, English, Hindi, Maithili

Selected Publications

See Google Scholar for a complete list.

[1] **Saurabh Kumar**, D. Mohanta, and P. K. Ghosh, *A Training Criterion with Token-Level Tolerance to Transcription Ambiguity for Automatic Speech Recognition*, Submitted to ICASSP 2027, 2026. arXiv: 2609.30160 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2609.30160.

[2] **Saurabh Kumar**, A. Singh, D. G, Amartyaveer, J. Bandekar, S. Murthy, S. Sharma, S. Badiger, S. Udupa, A. Nagireddi, S. Raghavan K M, R. Saxena, J. Nanavati, R. Nanavati, J. Sridharan, A. Mehta, A. Khuraishi K S, S. P. R. Mora, P. Venkataramakrishnan, G. Date, K. P, and P. K. Ghosh, "RESPIN-S1.0: A read speech corpus of 10000+ hours in dialects of nine Indian Languages", in *Advances in Neural Information Processing Systems*, vol. 38, Main Conference, Curran Associates, Inc., 2025. DOI: 10.52202/085713-4928. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/file/d88714d4f2c1e826d33d6c1eb1497ec9-Paper-Datasets_and_Benchmarks_Track.pdf.

[3] **Saurabh Kumar**, S. Sharma, G. Deekshitha, A. Singh, Amartyaveer, S. Udupa, S. Badiger, S. Khudanpur, S. Sitaram, S. Umesh, B. Ramabhadran, B. Kingsbury, H. A. Murthy, S. S. Narayanan, H. Lakougna, and P. K. Ghosh, "MADASR 2.0: Multi-Lingual Multi-Dialect ASR Challenge in 8 Indian Languages", in *Proc. IEEE ASRU*, 2025, pp. 1–7. DOI: 10.1109/ASRU65441.2025.11434729.

[4] **Saurabh Kumar**, Amartyaveer, and P. K. Ghosh, "Jointly Improving Dialect Identification and ASR in Indian Languages using Multimodal Feature Fusion", in *Proc. Interspeech 2025*, 2025, pp. 2770–2774. DOI: 10.21437/Interspeech.2025-421.

[5] Amartyaveer, **Saurabh Kumar**, S. Sharma, S. Udupa, S. Badiger, A. Singh, D. G, J. Bandekar, S. Murthy, and P. Kumar Ghosh, "Improving Dialect Identification in Indian Languages Using Multimodal Features from Dialect Informed ASR", in *Proc. ICASSP*, 2025, pp. 1–5. DOI: 10.1109/ICASSP49660.2025.10889099.

[6] S. Udupa, J. Bandekar, **Saurabh Kumar**, D. G, S. B, A. S, S. Murthy, P. Pai, S. Raghavan, R. Nanavati, and P. K. Ghosh, "Adapter pre-training for improved speech recognition in unseen domains using low resource adapter tuning of self-supervised models", in *Proc. Interspeech*, 2024, pp. 2529–2533. DOI: 10.21437/Interspeech.2024-1587.

[7] S. Udupa, J. Bandekar, G. Deekshitha, **Saurabh Kumar**, P. K. Ghosh, S. Badiger, A. Singh, S. Murthy, P. Pai, S. Raghavan, and R. Nanavati, "Gated Multi Encoders and Multitask Objectives for Dialectal Speech Recognition in Indian Languages", in *Proc. IEEE ASRU*, 2023, pp. 1–8. DOI: 10.1109/ASRU57964.2023.10389624.

[8] S. Udupa, J. Bandekar, A. Singh, D. G, **Saurabh Kumar**, S. Badiger, A. Nagireddi, R. R, P. K. Ghosh, H. A. Murthy, P. Kumar, K. Tokuda, M. Hasegawa-Johnson, and P. Olbrich, "LIMMITS'24: Multi-Speaker, Multi-Lingual INDIC TTS With Voice Cloning", *IEEE OJSP*, vol. 6, pp. 293–302, 2025. DOI: 10.1109/OJSP.2025.3531782.

[9] S. Pulikodan, A. Singh, A. Basu, N. Desai, P. K. J, P. D. Bhat, R. Dharmaraju, R. Gupta, S. Udupa, **Saurabh Kumar**, S. Sharma, V. Sanka, D. Tewari, H. Dhand, A. Kamat, S. Singh, S. Vashishth, P. Talukdar, R. Acharya, and P. K. Ghosh, "VAANI: Capturing the language landscape for an inclusive digital India", in *Advances in Neural Information Processing Systems (Evaluations and Datasets Track)*, To appear, 2026. arXiv: 2603.28714 [eess.AS]. [Online]. Available: https://arxiv.org/abs/2603.28714.

[10] S. Shahnawazuddin, A. Kumar, V. Kumar, **Saurabh Kumar**, and W. Ahmad, "Robust children's speech recognition in zero resource condition", *Applied Acoustics*, vol. 185, p. 108 382, 2022. DOI: 10.1016/j.apacoust.2021.108382.

[11] S. Shahnawazuddin, A. Kumar, **Saurabh Kumar**, and W. Ahmad, "Enhancing robustness of zero resource children's speech recognition system through bispectrum based front-end acoustic features", *Digital Signal Processing*, vol. 118, p. 103 226, 2021. DOI: 10.1016/j.dsp.2021.103226.